HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Department of Computer Science and Engineering
Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT 213 | **Course Title: Statistics and Probability**
Midterm Examination | Semester: Fall 2025
Date: 05/12/2025 | Total Time: 1:30 Hours | **Total Marks: 30**

Answer any three (03) out of the following questions: $3 \times 10 = 30$
[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.		Questions	CLOs	Marks
Q1.	(a)	Define continuous and discontinuous data sets with examples.	CLO1	[2]
	(b)	The scores (out of 100) obtained by 50 students in a mathematics test are as follows: 69, 48, 84, 58, 48, 73, 83, 48, 66, 58, 84,72, 40, 66, 64, 71, 64, 66, 69, 66, 83, 66, 69, 71,81, 71, 73, 69, 66, 66, 64, 58, 64, 69, 69, 33, 45, 52, 92, 82, 33, 25, 29, 42, 65, 72, 65, 67, 71, 70	CLO3	[4+4]

(i). Construct a grouped frequency distribution table for the given data and draw a histogram based on that table.

(ii). Classify the scores into grades (A+, A, B, C, D, F) based on your chosen grading scale, and represent the data with a bar graph showing the number of students in each grade.

No.		Questions	CLOs	Marks
Q2.	(a)	Define measures of central tendency with examples.	CLO1	[2]
	(b)	Using the step-deviation (short-cut) method, compute the arithmetic mean of the given data. Then determine both the geometric mean (G.M.) and the harmonic mean (H.M.) for the same dataset.	CLO3	[8]

$\frac{\sum f \log x_i}{N}$

Weight (in kg)	40 – 44	44 – 48	48 – 52	52 – 56	56–60	60–64
Frequency	5	3	13	8	6	3

No.		Questions	CLOs	Marks
Q3.	(a)	Define the measure of Skewness and Kurtosis.	CLO1	[2]
	(b)	The following are the marks of 150 students in an examination. Calculate (i) the skewness, (ii) the Kurtosis of this dataset and give your comment.	CLO3	[8]

Marks	0-10	10–20	20–30	30–40	40–50	50–60	60–70	70-80
Number of Students	10	40	20	0	10	40	16	14

No.		Questions	CLOs	Marks
Q4.	(a)	Define the measures of dispersion and their kinds. Why is the measure of dispersion better than measures of central tendency?	CLO1	[4]
	(b)	Find the quartile deviation and mean deviation from the following table. Give your comparative comment.	CLO3	[6]

Size	4-8	8-12	12–16	16–20	20–24	24–28	28–32	32-36	36-40
Frequency	6	10	18	30	15	12	10	6	2

$\frac{Q_3-Q_1}{2}$

$\frac{\sum f|x-\bar{x}|}{N}$

Cumulative

Q5. (a) The table below shows the distribution of marks for 100 students on a science test CLO3 [5]

Marks (%)	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90
Frequency	11	6	19	10	19	10	16	9

Find the variance and standard deviation from the following table. Give your comparative comment.

(b) Compute quartiles Q1, Q2, and Q3 from the grouped data and 90th percentile graphically. CLO3 [5]

Marks (%)	0 - 2	2 - 4	4 - 6	6 - 8	8 - 10	10 - 12	12- 14
Frequency	5	16	13	7	5	4	3

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Electrical and Electronic Engineering
CSE 213 : Digital Electronics

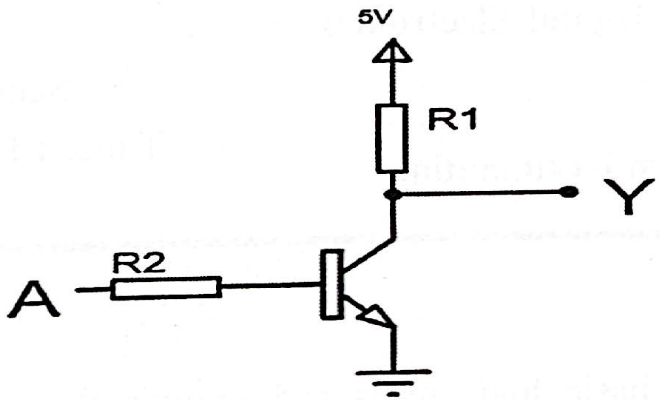
Batch: 27th | Semester: Fall 2025
Time: 1 Hour 30 Minutes
Date: 7/11/2025 | **Midterm Examination** | Total Marks: 30

Answer any five questions. $5 \times 6 = 30$

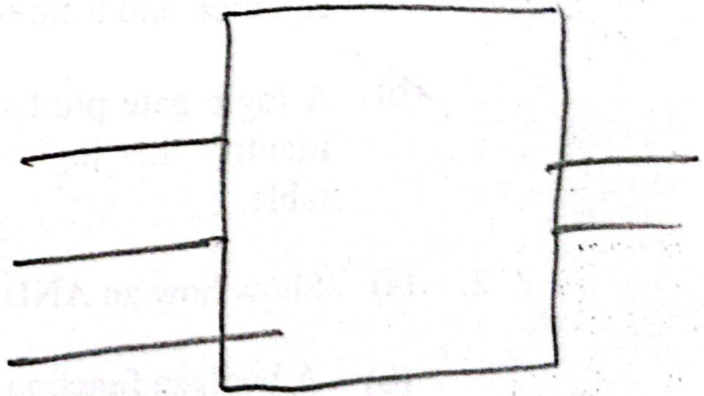
1.	(a)	Define logic gates. Describe the basic logic gates and provide their equation and truth table.	CLO1	3
	(b)	A logic gate produces a HIGH output when its two inputs are different. Identify this logic gate and describe its operation along with its truth table. X-NOR	CLO1	3
2.	(a)	Show how an AND gate can be implemented using only NAND gates.	CLO1	3
	(b)	A boolean function is given by: Y=AB+C i) Identify the basic gates required to implement the given function. ii) Implement the given function using only NOR gate.	CLO1	1+2=3
3.	(a)	A boolean function is defined as: Y=A+B i) Implement the given boolean function using diode logic. ii) Can the same logic be extended to three variables using diode logic? Justify your answer.	CLO3	1+2=3
	(b)	Is it possible to implement a NOT gate using diode logic? Explain your reasoning.	CLO3	3
4.	(a)	State the differences between a half adder and a full adder.	CLO1	2
	(b)	Describe full adder and implement a full adder using two half adders.	CLO1	4
5.	(a)	Define and describe the following terms with proper examples. i)SOP form ii)POS form ii)Canonical form	CLO1	3×1=3
	(b)	Simplify the following boolean function using a Karnaugh map (K-map): f(A,B,C,D)=∑m(0, 2, 3, 5, 7, 8, 10, 11, 14, 15)	CLO1	3
6.	(a)	Define the following terms: i) Encoder ii) Decoder	CLO1	2×1=2
	(b)	Describe a decoder circuit and implement a 8×3 decoder using basic logic gates.	CLO1	4

HAMDARD UNIVERSITY BANGLADESH

7. Observe the diagram and answer the following questions. CLO3 3×2=6



i) Identify the logic family used in the given diagram.
ii) Describe the given logic gate with truth table.
iii) Using the identified logic family, implement a 4-input AND gate.



HAMDARD UNIVERSITY BANGLADESH

Insertion
Deletion
Searching
Sorting

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE215: Data Structures

Batch: 27th | Semester: Fall 2025
Time: 1 hour 30 minutes
Date: 08/12/2025 | **Mid Term Examination** | Total Marks: 30

Answer any five questions from the following: 5x6= 30

1	(a) Define Data Structure and illustrate the fundamental operations performed on data structures with examples.	CLO1	3
	(b) Classify data structures into primitive and non-primitive types and summarize the purpose of each category with examples.	CLO1	3
2	(a) Describe a singly linked list and construct an algorithm for inserting a node at the beginning of the Singly Linked List.	CLO1	3
	(b) Given the list HEAD → 5 → 8 → 12 → 20 → NULL Perform the deletion of the node containing 12 and demonstrate the updated list.	CLO5	3
3	(a) Using Interpolation Search, calculate the probe positions and determine whether element 35 exists in A = [10, 20, 30, 40, 50, 60]	CLO2	3
	(b) Contrast interpolation search with binary search in terms of efficiency and usage scenarios.	CLO2	3
4	(a) Execute Selection Sort on [29, 10, 14, 37, 13] and display the array after each pass.	CLO2	3
	(b) Perform Insertion Sort on [22, 5, 11, 32, 4] and illustrate each intermediate step.	CLO2	3
5	(a) Using Radix Sort, sort the numbers [122, 45, 9, 802, 24] and illustrate bucket distribution after each pass.	CLO2	3
	(b) Evaluate the differences between Radix Sort and Counting Sort based on their design and performance characteristics.	CLO2	3
6	(a) Identify whether the following structures are linear or non-linear and justify your reasoning: Linked List, Binary Tree, Stack, Graph.	CLO1	2
	(b) Apply Merge Sort to [7, 2, 6, 3] and show the partitioning and merging process.	CLO2	4
7	(a) Examine why Ternary Search may not always be more efficient than Binary Search, despite dividing the list into three parts.	CLO4	2
	(b) Given the sorted list: A = [3, 7, 12, 18, 24, 31, 45, 56]. Perform Binary Search to locate 31, and illustrate the mid value updates.	CLO2	4

HAMDARD UNIVERSITY BANGLADESH

.Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT217
Course Title: Ordinary and Partial Differential Equations
Midterm Examination
Semester: Fall 2025
Date:14/12/2025
Total Time:1.50 Hours
Total Marks: 30

Answer any three (03) of the following questions: $3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.		Questions	CLOs	Marks
1.	(a)	Define ordinary differential equations, differential coefficient, order and degree of a differential equations.	CLO1	[3]
	(b)	Find the order and degree of the following differential equations: i. $\frac{d^3y}{dt^3} + 2(\frac{dy}{dt})^7 + 7y = 0$ ii. $\frac{d^5y}{dx^5} + (\frac{d^3y}{dx^3})^6 + 5y + 3 = 0$ iii. $\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} - 2y = 0$	CLO3	[2]
	(c)	Find the differential equation of the form: $y = e^{-x}(Acosx + Bsinx)$	CLO3	[3]
	(d)	Solve the differential equation: $x\sqrt{1-y^2}dx + y\sqrt{1-x^2}dy = 0$.		[2]
2.	(a)	What is homogeneous differential equation?	CLO1	[1]
	(b)	Solve the homogeneous differential equation: $(x^2 + y^2)dx - 2xydy = 0$.	CLO3	[3]
	(c)	Solve the following differential equation by separation variable: i. $Sin^{-1}\left(\frac{dy}{dx}\right) = x + y$. ii. $x\frac{dy}{dx} - y = x\sqrt{x^2 + y^2}$.	CLO3	[6]
3.	(a)	What is exact differential equation?	CLO1	[1]
	(b)	Prove that the necessary and sufficient condition for a differential equation of first degree being exact.	CLO2	[4]
	(c)	Solve the following differential equations: i. $(x^3 + 3xy^2)dx + (y^3 + 3x^2y)dy = 0$. ii. $(1 - x^2)\frac{dy}{dx} - xy = 1$.	CLO3	[5]
4.	(a)	Define linear differential equations.	CLO1	[1]
	(b)	Solve the differential equation with variable coefficients: $x^3\frac{d^3y}{dx^3} + 3x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = x + x^2lnx$.	CLO3	[6]
	(c)	Solve the differential equation: $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = x^2e^{2x}$.	CLO3	[3]
5.	(a)	Solve the following equation by solvable for x : $y = yp^2 + 2px$.	CLO3	[5]
	(b)	Solve the following equation solvable for y : $y = 2px + y^2p^3$.	CLO3	[5]

Good Luck

HAMDARD UNIVERSITY BANGLADESH

Department of CSE
Faculty of Science, Engineering and Technology
2nd Year 3rd Semester Examination

Course Code: BAN211 Course Title: History of Emergence of Independent Bangladesh
Midterm Examination Semester: Fall 2025
Date: 12/11/2025 Total Time: 1.5 Hours Total Marks: 30

[The figure in the right margin indicates full marks]

[Answer all of the following]

No.	Question	Marks	Level	CLOs
1.	Discuss the naming of Bengal and the background of the Muslim conquest of Bengal.	10	Understand	1
2.	Write the biography of Shamsuddin Iliyas Shah.	10	Remember	2
3.	Describe the borders of Bengal during the Muslim Sultanate.	10	Understand	1

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

Program: BSc in Computer Science and Engineering

Mid-Term Examination- Fall 2025

Course: ACT-215: Financial and Managerial Accounting

Total Marks: 30 **Time: 1 Hour 30 Minutes**

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answer to questions should be neat and clean.

No.	Question/ Statement	CLOs	Level	Marks
1.	Answer any <u>two</u> of the following questions; (a)Define accounting? Briefly describe the users of accounting. (b)Explain the Historical cost principles and the business entity assumption. (c). "Bookkeeping and accounting are the same." Do you agree? Explain.	CLO 1	Understanding	10
2	(a). Describe the basic steps in the recording process. (b). Hillen Meyer was started on April 1,2025 by F. L. Wright and associates. The following selected events and transactions occurred during April. Apr. 1. Hillen Meyer invested $35,000 cash in the business. 4. Purchased land costing $27,000 for cash. 8. Incurred advertising expense of $1,800 on account. 11. Paid salaries to employees $1,500. 12. Hired Park manager at a salary of $4,000 per month, effective May 1. 13. Paid $1,650 cash for a one-year insurance policy. 17. Withdrew $1,000 cash for personal use. 20. Received $6,800 in cash for admission fees. 25. Sold 100 coupon books for $25 each. Each book contains 10 coupons that entitle the holder to one admission to the park. 30. Received $8,900 in cash admission fees. 30. Paid $900 on balance owed for advertising incurred on April 8. Instructions; Journalize the April transactions	CLO 2	Analyzing,	10
3.	(a) Indicate whether each of the statements presented below is transactions or events. 1) Depreciation of a company vehicle over time tk.5,000 2) A customer placing an order for goods which value tk.1,500. 3) An owner withdrawing cash for personal use tk.500. 4) Owner met with supplier for purchasing materials which worth tk.1,50,000 5) Sold old printer for tk.3,000 6) Signing a major contract with a new client 7) Bought stock worth tk.25,000 8) Paid rent of tk.15,000 9) Paying an employee's salary tk.30,000 10) A competitor launching a new product	CLO 3	Applying	5

(b). Presented below is selected information related to Rivera Company SA at December 31, 2025. Rivera reports financial information monthly.

Accounts Payable	Tk3,000	Rent Expense	9,800
Salaries and Wages Expense	16,500	Service Revenue	54,000
Cash	9,000	Accounts Receivable	13,500
Notes Payable	25,000	Equipment	29,000
Advertising Expense	6,000	Drawing	7,500

Instructions;

Determine the net income that Rivera Company SA reported for December 2025

CLO 3 | Applying | 5

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4 (a) Explain Forensic Accounting.

(b)Kinney's Repair Ltd. was started a business on May 1,2025. A summary of May transactions is presented below.

1. Owner's invested tk.10,000 cash in the business.
2. Purchased equipment for tk.5,000 cash.
3. Paid tk.400 cash for May office rent.
4. Paid tk.700 cash for supplies.
5. Incurred tk.250 of advertising costs in the Beacon News on account.
6. Received tk.4,700 in cash from customers for repair service.
7. Owner's withdraw tk.1,000 cash for personal use
8. Paid part-time employee salaries tk.1,000.
9. Paid utility bills tk.160.
10. Performed repair service worth tk.980 on account.
11. Collected cash of tk.120 for services billed in transaction (10).

Instructions

Prepare a tabular analysis of the transactions, using the following column headings: Cash, Accounts Receivable, Supplies, Equipment, Accounts Payable, Capital, Revenues, Expenses, and Drawing.

CLO 4 | Understanding . Evaluating | 10

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5 Mr. Subhas Das is a licensed FCA and IT Practitioner. During the first month of operations of his business, the following event and transactions are occurred.

May 1 Mr. Das invested in cash Tk. 70,000.

2. Hired a secretary-receptionist at a salary of Tk. 5,000 per month.

3. Purchased Tk. 15,000 of supplies on account from Acme Supply Company.

7. Paid office rent of Tk. 9,000 in cash for the month.

11. Completed a tax assignment and billed client Tk. 20,100 for services provided. A/R

12. Received Tk. 30,500 advance on a management consulting engagement.

17. Received cash of Tk. 10,200 for services completed for MGH Group and Co.

30. Paid secretary-receptionist Tk. 5,000 for the month.

31 Paid Tk. 9000 of balance due Acme Supply Company.

Instructions:

(a) Journalize the May transactions.

(b) Post to the ledger accounts. .

CLO 4 | Evaluating | 10

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Department of Computer Science and Engineering
Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

15.31

Course Code: MAT 213 | **Course Title: Statistics and Probability**

Final Examination | Semester: Fall 2025

Date: 01/03/2026 | Total Time: 2:00 Hours | **Total Marks: 40**

Answer any four (04) out of the following questions: $4\times 10 = 40$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.		Questions	CLOs	Marks
Q1.	(a)	Prove that the value of correlation coefficient lies between -1 to +1.	CLO2	[3]
	(b)	Study is conducted involving 14 infants to investigate the association between gestational age at birth, measured in weeks, and birth weight, measured in grams, and their Gestational age and their Weight at birth are recorded below:	CLO3	[7]

Infant no.	1	2	3	4	5	6	7	8	9	10
Gestational age	34.7	36	29.3	40.1	35.7	42.4	40.3	37.3	40.9	38.3
Birth Weight	1895	2023	1440	2835	3090	3827	3260	2690	3285	2920

Applying the proper method; Estimate the association between gestational age and infant birth weight.

Q2. (a) A data analyst working for a climate research institute is studying whether urban air temperature can be statistically explained by traffic density in a metropolitan area. Measurements were taken on 10 randomly selected days during the summer. The analyst proposes a linear statistical model to investigate the relationship. — CLO3 [10]

Where $x \in [5,30]$, where x is a random odd variable.

$$Y = 27 + 0.18X$$

X = Average daily traffic density (thousands of vehicles/days)

Y = Average daily air temperature (°C)

(i). Why does a strong correlation not necessarily imply that traffic density causes temperature changes.

(ii). The real temperature follows the above relationship. Compute and interpret the coefficient of determination R^2.

Q3. (a) Explain the properties and applications of the t-distribution and chi-square distribution. — CLO1 [5]

(b) Certain pesticide is packed into bags by a machine. A random sample of 10 bags is drawn and their contents are found to weigh (in kg) as follows: — CLO3 [5]

50 49 52 44 45 48 46 45 49 45

Test if the average packing can be taken to be 50 kg.

Q4. (a) Define the Chi-square (χ^2) test statistic and discuss the properties of the Chi-square distribution. CLO2 [4]

(b) Two researchers adopted different sampling techniques while investigating the same group of students to find the number of students falling in different intelligence levels. The results are as follows. CLO3 [6]

Researchers	No. of students				Total
	Below average	Average	Above average	Genius	
X	86	60	44	10	200
Y	40	33	25	2	100
Total	126	93	69	12	300

Would you say that the sampling techniques adopted by the two researchers are independent?

Q5. (a) What are the components of time series analysis? Explain each component briefly. CLO1 [4]

(b) The following are the annual profits in thousand of taka in a certain business. Use the method of lest squares to determine the trend value of profit. Also forecast the annual profits for 2009. CLO3 [6]

Year	Profit
1999	60
2000	72
2001	75
2002	65
2003	80

Year	Profit
2004	85
2005	95
2006	101
2007	107
2008	115

Q6. (a) Define marginal probability. CLO1 [2]

(b) Mr. Ali feels that he will pass math 2/3 and stat 5/6 if the probability that he will pass at least one of the courses. CLO3 [4]

(c) Three contractors A, B, and C are bidding for the construction of a new academic building of NSTU. On the basis of previous experience and expert opinion, it is known that: Contractor A has exactly half the chance of B of winning the contract. Contractor B has a chance that is 4/3times the chance of C. If only one contractor is to be awarded the contract, find the probability that each contractor will get the contract. CLO3 [4]

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE213: Digital Electronics

Batch: 27th

Date: 8-3-2026

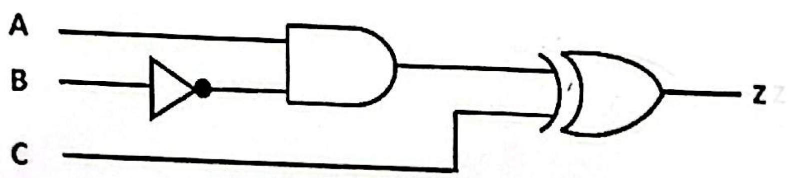
Final Examination

Semester: Fall 2025
Time: 2 Hours
Total Marks: 40

Answer any five questions of the followings. $5 \times 8 = 40$

1. a) What is propagation delay? How does propagation delay affect logic gate outputs? — CLO1 — 2

   b) Using the given logic circuit answer the following question. — CLO1 — 6



   All the logic gates shown in the logic circuit has a propagation delay of 20 ns. For time t<0, A=~~B~~=C=0 and B=1. At time t=0, all the inputs are flipped and remain in that state. Find out the duration when Z=1.

4+1

2. a) What is a multivibrator? Explain different types of multivibrators with proper example. — CLO4 — 4

   b) Differentiate between latch and flip flop. — CLO4 — 4

3. a) Do you support the statement "D flip flop have a invalid state"? Justify your answer. — CLO4 — 2

6+1

   b) Explain SR flip flop with all the necessary tables and equations. — CLO4 — 6

4. a) Design a 2-bit asynchronous up counter. — CLO4 — 2

   b) Explain race condition in JK flip flop with proper timing diagram. Describe the remedy for the race condition. — CLO4 — 6

5. a) Design a 3-bit asynchronous down counter. — CLO4 — 3

   b) Design a 3-bit asynchronous up-down counter. — CLO4 — 5

6 6. Explain JK flip flop with all the necessary tables and equations. — CLO4 — 8

7. Design a 3-bit synchronous down counter. — CLO4 — 8

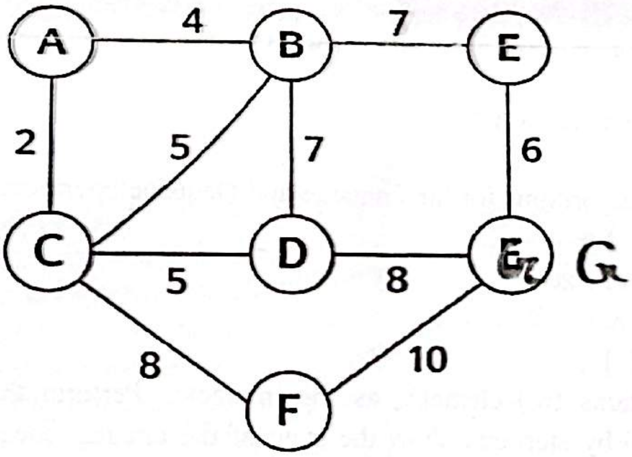
HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Mathematics
CSE215: Data Structure

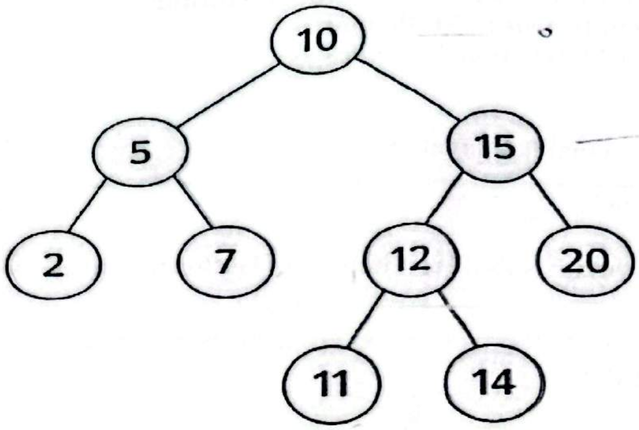
Date: 03/03/2026 **Final Examination** Semester: Fall 2025
Time: 2 hours
Total Marks: 40

Answer any **five** questions from the following: 5x8= 40

1	(a) Define a weighted-2 tree. Name two real-life applications of Huffman coding in data compression.	CLO 1	3
	(b) Construct a Huffman Tree for the following string and provide the resulting encoded string: "BEEP_BOP_BOOP".	CLO 4	5
2	(a) Convert the following infix expression to a postfix expression using a Stack-based algorithm: A + B * (C - D) / E.	CLO 2	6
	(b) Convert the following Arithmetic Expression (10 + 5) * 2 – (12 / 3) to a Binary Tree.	CLO1	2
3	(a) Define a Minimum Spanning Tree (MST). Explain its properties.	CLO 1	3
	(b) Using Prim's Algorithm, find the MST for a graph with 5 vertices. Explain the importance of the starting vertex in this algorithm.	CLO 4	5



4	(a) For the given a binary tree, find the In-Order, Pre-order and Post-order traversal.	CLO 1	3



	(b) Find the following terms for the above tree: Level, Depth, Terminal Nodes, Nodes with 2 successors, and Nodes with 1 successors.	CLO 4	5
5	(a) Perform the insertion of the following keys into an initially empty BST: {45, 10, 7, 90, 12, 50}. Show the final tree.	CLO 1	4
	(b) Analyze the time complexity for inserting and deleting a node in a BST for both the average and worst-case scenarios.	CLO 4	4
6	(a) Define a Max-Heap and a Min-Heap. Discuss the variants of the Heap data structure.	CLO 1	4
	(b) Convert the array [40, 18, 35, 12, 25, 10, 30] into a Min Heap using the heapify algorithm. Show all intermediate steps and draw the final heap.	CLO 4	4
7	(a) Write the pseudocode or algorithms for the Enqueue and Dequeue operations in a Circular Queue.	CLO 3	4
	(b) Consider a circular queue of size 5.	CLO 5	4

Index: 0 1 2 3 4

Queue: [10] [20] [ ] [ ] [ ]

Initially, the queue contains two elements as shown above. Perform the following operations step by step and show the state of the circular queue after each operation:

i. Enqueue(30)
ii. Enqueue(40)
iii. Dequeue()
iv. Enqueue(50)
v. Enqueue(60)
vi. Dequeue()

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
B.Sc. (honors) in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT217 | Course Title: Ordinary and Partial Differential Equations
Final Examination | Semester: Fall 2025
Date: 03/12/2026 | Total Time:2:00 hours | **Total Marks: 40**

Answer any four (04) of the following questions: $4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

1. **(a)** Define linear and nonlinear partial differential equations with examples. CLO1 [2]

   (b) Define order and degree of a partial differential equation. Find the order and degree of the following partial differential equations: CLO1 CLO2 [2]

   (i) $\frac{\partial^6 z}{\partial^6 x} + \left(\frac{\partial z}{\partial y}\right)^2 = z.$

   (ii) $\left(\frac{\partial^2 z}{\partial^2 x}\right)^{14} + x^5 \left(\frac{\partial z}{\partial y}\right)^5 = 2.$

   (c) Find the general solution of the differential equation by eliminating arbitrary constants: $z = Ae^{-\alpha t} cos\beta x \; sin\gamma y.$ $[\alpha^2+\beta^2+\gamma^2=0] \to$ Condition CLO3 [3]

   (d) Find the general solution of the differential equation by eliminating arbitrary functions : $x + y + z = f(x^2 + y^2 + z^2)$ CLO3 [3]

2. **(a)** State and proof the solution of Lagrange method for partial differential equations. CLO2 [5]

   (b) Find the general solution of the following equations by Lagrange method: CLO3 [5]

   (i) $y^2 zp - yx^2 zq = x^2 y$

   (ii) $(x - y)p + (x + y)q = z(y - x)$

3. **(a)** What do you mean by surface integral? Provide a brief introductions. CLO1 [1]

   (b) Find the integral surface of the of the PDE $(x - y)p + (y - x - z)q = z$, which passes through the line $x^2 + y^2 = 1, z = 1$. CLO3 [4]

   (c) Find the complete and singular integral of : $2xz - px^2 - 2qxy + pq = 0$ CLO3 [5]

4. **(a)** What is complete integral for PDE? CLO1 [1]

   (b) Derive Charpit's method for the solution of a first order non-linear partial differential equation $f(x, y, z, p, q) = 0$. CLO2 [5]

   (c) Apply Charpit's method to solve the following PDE; CLO3 [4]

   (i) $px + qy = pq.$

   (ii) $pxy + pq + qy - yz = 0.$

5. **(a)** Solve the following PDE: CLO3 [6]

   (i) $2r + 5s + 2t = 0.$

   (ii) $(2D^2 - 5DD' + 2D'^2)z = 2x + 3y$

   (iii) $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 2\frac{\partial^2 z}{\partial y^2} = (2x + y)^{\frac{1}{2}}.$

   (b) Discuss the nature of the equations $u_{xx} + (x - 1)u_{xy} + u_{yy} - 2x^2 u_x + 3xyu_y + 2u = sinx.$ CLO3 [4]

6. **(a)** Define boundary problems, and how many methods to solve boundary value problems. CLO2 [2]

   (b) Solve the boundary value problem $\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}; 0 < x < l;\ t > 0$, Using the conditions: $u(0, t) = u(l, t) = 0; u(x, 0) = f(x).$ CLO3 [8]

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Department of CSE
Faculty of Science, Engineering and Technology

Course Code: BAN211 **Course Title: History of Emergence of Bangladesh**

Final Examination Semester: Fall 2025

Date: 3/5/2026 Total Time: 2.00 Hours **Total Marks: 40**

[The figure in the right margin indicates full marks]

[Answer all of the following]

No.	Question	Marks	Level	CLOs
1.	How many parliamentary elections have been held in Bangladesh from 1973 to 2026, and who formed the government? Discuss elaborately.	14	Analysis	4
2.	Write the names of the three major organs of the government and discuss their functions. OR Explain elaborately the ways to prevent violence in Bangladesh's electoral system.	13	Remember Understand	3
3.	Describe the role of the upper house in the governance system of Bangladesh. OR Discuss the amendments to the Constitution of Bangladesh.	13	Understand Remember	2

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
Program: **BSc in Computer Science and Engineering**
Final Examination- Fall 2025
Course: ACT-215: Financial and Managerial Accounting

Total Marks: 40 **Time: 2 Hour**

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answer to questions should be neat and clean.

No.	Question/ Statement	CLOs	Marks
1.	What are the three major elements of product costs in a manufacturing company? Or, Define accrual basis and cash basis accounting.	4	5
2.	Differentiate between a product cost and a period cost. Or, A variable cost is a cost that varies per unit of product, whereas a fixed cost is constant per unit of product." Do you agree? Explain	6	7
3.	The Meghna Lawn care company Ltd. began on April 1. At April 30, the trial balance shows the following balances for selected accounts. (Amounts are in Taka)	4, 6	8

Prepaid Insurance	3,600
Equipment	28,000
Notes Payable	20,000
Unearned Service Revenue	4,200
Service Revenue	1,800

Analysis reveals the following additional data.

1. Prepaid insurance is the cost of a 2-year insurance policy, effective April 1
2. Depreciation on the equipment is Tk.500 per month.
3. The note payable is dated April 1. It is a 6-month, 12% note.
4. Seven customers paid for the company's 6-month lawn service package of Tk.600 beginning in April. The company performed services for these customers in April.
5. Lawn services performed for other customers but not recorded at April 30 totaled Tk.1,500.

Instructions:
Prepare the adjusting entries for the month of April. Show computations.

Or,
The following cost and inventory data are taken from the accounting records of mason company for the year just completed.

Costs incurred	Taka
Advertising expense	90,000
Sales commissions	50,000
Direct labor cost	70,000
Purchases of raw materials	1,18,000
Manufacturing Overhead	80,000
Depreciation, office equipment	3,000

Inventories;	Beginning of Year	End of Year
Raw materials	7,000	15,000
Work in process	10,000	5,000
Finished goods	20,000	35,000

Required:
Prepare a schedule of the cost of goods manufactured.

4. At the end of its first month of operations, Pampered Pet Service Ltd. has the following unadjusted trial balance 4,6 15

	Debit	Credit
Cash	5,400	
Accounts Receivable	2,800	
Supplies	1,300	
Prepaid Insurance	2,400	
Equipment	60,000	
Notes Payable		40,000
Accounts Payable		2,400
Service Revenue		4,900
Capital		30,000
Owners Drawings	1,000	
Utilities Expense	800	
Salaries and Wages Expense	3,200	
Advertising Expense	400	
	77,300	77,300

Other data:

1. Insurance expires at the rate of TK.200 per month.
2. TK.1,000 of supplies are on hand at August 31.
3. Monthly depreciation on the equipment is TK.900.
4. Interest of TK.500 on the note's payable has accrued during August.

Instructions:

(a)Prepare the adjusting entries. (b)Complete the worksheet.

Or,

The following cost and inventory data for the just completed year are taken from the accounting records of Eccles Company

Costs incurred	Taka
Advertising expense	1,00,000
Sales commissions	35,000
Direct labor cost	90,000
Purchases of raw materials	1,32,000
Rent, factory building	80,000
Indirect labor, factory	56,300
Utilities, factory	9,000
Supplies, factory	700
Maintenance, factory equipment	24,000
Depreciation, office equipment	8,000
Depreciation, factory equipment	40,000

Inventories;	Beginning of Year	End of Year
Raw materials	8,000	10,000
Work in process	5,000	20,000
Finished goods	70,000	25,000

Required:

1.Prepare a schedule of cost of goods manufactured and prepare a schedule of cost of goods Sold.

5 Discuss the financial reporting concepts. 4 5

Or,

The Income statement of Hamdard laboratories Ltd. for the month of July shows net income of TK.4,000 based on Service Revenue TK.8,700, Salaries and Wages Expense TK.2,500, Supplies Expense TK.1,700, and Utilities Expense TK.500. In reviewing the statement, you discover the following.

1. Insurance expired during July of TK.700 was omitted.
2. Supplies expense includes TK.250 of supplies that are still on hand at July 31.
3. Depreciation on equipment of TK.300 was omitted.
4. Accrued but unpaid salaries and wages at July 31 of TK.400 were not included.
5. Services performed but unrecorded totaled TK.650.

Instructions:

Determine a correct income statement for July 2025.